Somya Sagarika Kar

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OBJECTIVE

Enthusiastic Data Science postgraduate with practical experience in AI/ML and deep learning projects, including multimodal sarcasm detection. Seeking an entry-level role to apply machine learning, data analysis, and problem- solving skills in building intelligent systems.

EDUCATION

Silicon University ,Bhubaneswar Odisha, India
Master in Data Science 2024 – 2026
CGPA 8.68

INTERNSHIPS & TRAINING

Devops Masters Intern- Ingenious TechWorld **2025**

- Provisioned and managed AWS cloud infrastructure (EC2, S3, IAM, VPC) and containerized applications using Docker and Kubernetes, implementing Horizontal Pod Autoscaling and Ingress controllers for scalable deployments.
- Built end-to-end CI/CD pipelines using GitHub Actions and AWS CodeBuild with ArgoCD for GitOps-based deployment, and automated infrastructure provisioning using Terraform (IaC); monitored system performance via Grafana and Prometheus dashboards.

CERTIFICATIONS

• Introduction To Industry 4.0 And Industrial Internet Of Things - NPTEL
• Python (Basic, Advance) - Hackerrank
• AI & Machine Learning – Microsoft Learn

PROJECTS

Sarcasm Detection in Text and Image Using Deep Learning Techniques **September 2025**

- Built a **Multimodal Sarcasm Detection** system using deep learning (Bi-LSTM + GloVe + ResNet50 with Global Average Pooling), fusing Text, Image, and Attribute modalities via a Hierarchical Attention Fusion architecture on a 21,586-tweet dataset.
- Proposed Hierarchical Attention Model achieved **83% accuracy and 0.81 precision**, outperforming the Simple Fusion baseline (82% accuracy, 0.78 precision), with an ROC-AUC of 0.90 — validating dual attention mechanisms for resolving cross-modal semantic incongruity.
- Implemented data sanitization to remove ground-truth leakage keywords, applied GloVe (200D) embeddings with sequence padding, and adapted ResNet50 with GAP to generate compact 2048-dimensional visual feature vectors for computationally efficient multimodal fusion.

Amazon recommendation system **April 2025- May 2025**

- Built a personalized recommendation system on the Amazon product dataset (10K reviews) using collaborative filtering and matrix factorization (Scikit-learn, Python), achieving improved prediction accuracy through hyperparameter tuning and evaluated model performance using RMSE and precision k metrics.

CodePipeline using Terraform & DevSecOps with GitHub Actions and Kubernetes Sealed Secrets **June 2025**

- Provisioned AWS CodePipeline using Terraform IaC, integrated DevSecOps security scanning (tfsec, Trivy) via GitHub Actions to detect misconfigurations, and managed Kubernetes Sealed Secrets for encrypted secret handling across namespaces in a GitOps-based deployment workflow.

TECHNICAL SKILLS

Programming Languages: C, Python(pandas, NumPy, Scikit-learn), SQL
Frameworks: Jupyter, VS Code, Bootstrap
Web Development: HTML, CSS, Javascript
Core Concepts: DSA , OOPS , DBMS , CN ,OS
Data Science: Machine Learning, NLP, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn